

Multi-View Fusion, CrossView Transformer, and Uncertainty-Aware Triage for Breast Cancer Detection on VinDr-Mammo

From 0.756 to 0.846 AUC

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Background: Breast cancer is the most commonly diagnosed cancer worldwide. Population screening based on full-field digital mammography (FFDM) reduces mortality but imposes heavy radiologist workload and substantial inter-reader variability, motivating computer-aided diagnosis (CAD).

Purpose: To develop and evaluate a six-stage pipeline for breast cancer detection on the VinDr-Mammo FFDM benchmark, progressing from single-view CNN baselines to a multi-view ensemble with uncertainty-aware triage simulation.

Materials and Methods: Five single-view CNN architectures (Stage 1), BreastPairNet shared-backbone multi-view network across twelve variants (Stage 2), CrossView Transformer with spatial cross-attention (Stage 3), bounding-box annotation ablation (Stage 3b), MC Dropout uncertainty quantification (Stage 4), 16-model ensemble with TTA (Stage 5), and Verboom-style hybrid triage simulation (Stage 6). External validation on RSNA 2022.

Results: DenseNet-121 attained AUC=0.7564 (Stage 1). BreastPairNet improved to AUC=0.8256 (+9.1%, Stage 2); CrossView Transformer reached AUC=0.8317 (+10.0%, Stage 3). Bounding-box annotations degraded AUC by 0.12–0.20. The 16-model ensemble achieved AUC=0.8458 (+11.8%) with S@90%spec=0.6566. Hybrid triage at 50% AI reads preserved 87.9% sensitivity. External RSNA 2022 AUC=0.6008.

Conclusion: A six-stage multi-view pipeline achieves AUC=0.8458 on VinDr-Mammo. Bounding-box shortcut learning is a key negative finding; domain generalisation to US FFDM remains an open challenge.

Keywords: breast cancer detection; mammography; multi-view learning; uncertainty quantification; triage; VinDr-Mammo.

Key Results at a Glance

S@90%spec on VinDr-Mammo: **0.6566** (best ensemble) vs 0.1636 (RSNA 2022 — large domain gap).

1 Introduction

BREAST cancer is the most commonly diagnosed cancer worldwide and a leading cause of cancer-related death in women. Population screening programmes based on full-field digital mammography (FFDM) have repeatedly been shown to reduce breast-cancer mortality, yet they impose a heavy operational burden: in many countries each screening exam is double-read by two radiologists, and reader sensitivity varies by 10–30 percentage points between individuals [10, 11]. False-negative rates of 10–30% and false-positive recall rates of approximately 8% are well documented across both clinical practice and large CAD literature, with the historical, hand-engineered CAD systems failing to deliver measurable benefit in routine US screening [17].

Deep learning has dramatically improved the situation. CNNs trained on hundreds of thousands of exams now reach or exceed individual radiologist AUC on inter-

nal and some external cohorts [11], and meta-analyses report pooled AUC of 0.87 for AI on digital mammography versus 0.81 for radiologists [4]. Three trends are particularly relevant. First, multi-view modelling that fuses the craniocaudal (CC) and mediolateral oblique (MLO) projections improves over single-view inference because findings often appear unambiguously in only one view [10]. Second, transformer-based fusion of views allows feature-level interaction rather than score-level averaging. Third, uncertainty-aware inference via MC Dropout enables a *hybrid triage* protocol in which the AI reads the most-confident exams autonomously and refers the rest to a human reader [12], reducing radiologist workload by approximately 38% in retrospective evaluation [12].

The VinDr-Mammo benchmark [1] is a 20,000-image, patient-level, 4-view FFDM dataset. Three gaps in the literature are evident. (i) **Single-view dominance.** Most published VinDr-Mammo results train one image at a time. (ii) **Spatial cross-view attention is underexplored.** Existing multi-view designs typically concatenate global features. (iii) **Uncertainty**

Table 1: Summary of the main AUC results across the six-stage pipeline.

Stage	Method	AUC	vs. single-view
Prior work	MamT4 [18] (4-view Transformer)‡	0.840	—
Stage 1	DenseNet-121 (best single-view)	0.756	baseline
Stage 2	EfficientNet-B3 +reg (BreastPairNet)	0.826	+9.1%
Stage 3	ConvNeXt-Base (CrossView Transformer)	0.832	+10.0%
Stage 4	DenseNet-121/concat MC Dropout	0.810	sensitivity ↑
Stage 5	16-model ensemble + TTA×5	0.846	+11.8% *
External	RSNA 2022 (zero-shot transfer)	0.601	domain gap !

Input: 20,000 VinDr-Mammo DICOM images from 5,000 patients.

Preprocessing: DICOM decoding, percentile clipping, CLAHE, orientation normalisation.

Stage 1: single-view CNN baselines, best AUC = 0.7564.

Stage 2: BreastPairNet CC+MLO fusion, best AUC = 0.8256.

Stage 3: CrossView Transformer, best AUC = 0.8317.

Stage 3b: annotation ablation, all painted-box variants lose 0.122–0.201 AUC.

Stage 4: MC Dropout, entropy-ranked uncertainty estimates.

Stage 5: 16-model ensemble with asymmetry features and TTA, best AUC = 0.8458.

Stage 6: hybrid triage simulation and external RSNA 2022 validation.

Figure 1: Overview of the six-stage breast cancer detection pipeline.

quantification and triage simulation are rare. No published study has applied MC Dropout to VinDr-Mammo and simulated a Verboom-style triage policy.

This paper makes seven contributions: (1) systematic single-view CNN comparison; (2) **BreastPairNet** multi-view network with twelve backbone×fusion×resolution variants; (3) **CrossView Transformer** with spatial cross-attention; (4) **bounding-box annotation ablation** revealing a negative result; (5) **16-model ensemble** reaching AUC=0.8458; (6) **MC Dropout uncertainty calibration**; (7) **hybrid triage simulation** demonstrating 50% workload reduction.

Figure 1 illustrates the full pipeline.

2 Related Work

Deep CNN backbones. Residual networks [7] and densely connected networks [8] underpin almost all modern CAD systems. CheXNet [6] established DenseNet-121 as a default starting point for chest imaging. EfficientNet [9] introduced compound scaling (B3: 12M parameters). ConvNeXt [16] modernised pure-convolutional design to transformer-level ImageNet accuracy.

Multi-view mammography. Wu et al. [10] trained a four-stream ResNet-22 on 229,426 NYU exams (AUC=0.895), establishing that shared backbones and feature-level aggregation are essential under limited data. McKinney et al. [11] reported AUC=0.889 on UK FFDM, outperforming six radiologists. Cross-view spatial attention outperforms score-level averaging, motivating Stage 3.

VinDr-Mammo-specific work. Nguyen et al. [1] released VinDr-Mammo with BI-RADS, density, and lesion-level annotations. Ibragimov et al. [18] proposed

MamT4 (AUC=0.840±0.017). Abdikenov et al. [2] demonstrated CLAHE+YOLOv12 mAP gains; Mellado et al. [3] showed competitive patch-level F1 under limited annotation.

Class imbalance and focal loss. Focal loss [5] suppresses easy negatives via $(1 - p)^\gamma$ down-weighting; Johnson & Khoshgoftaar [15] survey algorithm-level imbalance methods.

Uncertainty quantification. Kendall and Gal [13] formalised aleatoric and epistemic uncertainty; MC Dropout provides tractable Bayesian model averaging. Verboom et al. [12] demonstrated 38.1% workload reduction on 41,469 Dutch screening exams while preserving cancer detection rate.

3 Dataset

3.1 VinDr-Mammo (primary dataset)

VinDr-Mammo [1] contains 20,000 FFDM images from 5,000 patients, four views per study (R-CC, L-CC, R-MLO, L-MLO) stored as DICOM, with BI-RADS (1–5), density (A–D), and lesion-level bounding box annotations.

Table 2: BI-RADS distribution in VinDr-Mammo and binary label mapping.

BI-RADS	Images	Binary label
1	14,036	negative
2	4,144	negative
3	832	negative
4	824	positive
5	164	positive
Total	20,000	988 positive (4.94%)

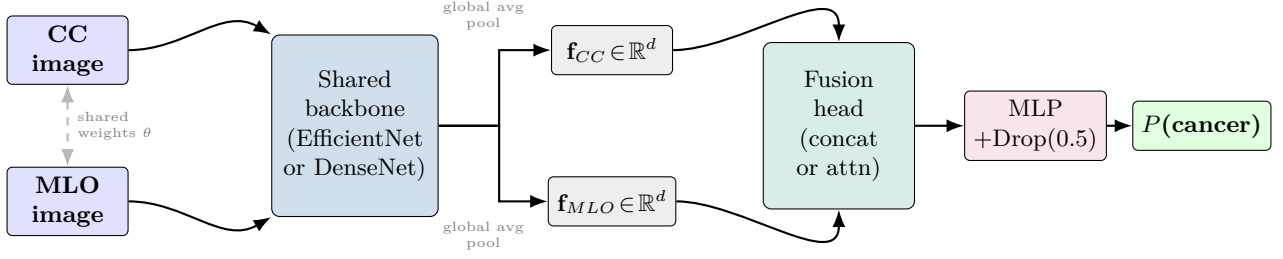


Figure 2: BreastPairNet architecture (Stage 2). CC and MLO images from the same breast pass through a *shared* CNN backbone (weights θ are tied for both views, indicated by the dashed arrow). Each view produces a global-average-pooled feature vector; the two vectors are fused by either concatenation (dim $2d$) or learned attention weights, then classified by a two-layer MLP with 50% dropout.

The official split is 4,000 training patients (16,000 images) and 1,000 test patients (4,000 images). For multi-view experiments we construct one ipsilateral CC+MLO pair per breast, yielding 8,000 training pairs and 2,000 test pairs, of which 99 test pairs (4.95%) are cancer-positive. A 5% validation slice (~ 400 pairs) is held out for early stopping; the official test split is never used for model selection.

3.2 RSNA 2022 (external dataset)

The RSNA 2022 dataset comprises 54,706 screening DICOMs from 11,913 patients. After 8-bit normalisation identical to VinDr-Mammo preprocessing, approximately 29,519 images required `pylibjpeg` to decode JPEG Lossless DICOMs. Non-CC/MLO views were excluded. The final paired dataset contains 23,826 CC+MLO pairs (19,062 training, 4,764 validation) on an 80/20 patient-level split.

4 Methods

4.1 DICOM preprocessing (12/14-bit $\rightarrow$ 8-bit PNG)

Four-step pipeline: (1) decode with `pydicom+pylibjpeg`; (2) percentile-clip to $[1\%, 99\%]$; (3) CLAHE [14] (32×32 tiles, clip-limit 2.0); (4) standardise orientation via horizontal flip when left half is brighter. Output: single-channel 8-bit PNG replicated to 3 channels.

4.2 Single-view CNN backbones

Five configurations: **VGG19**, **ResNet-50** (224 px), **ResNet-50** (384 px), **EfficientNet-B3** (224 px), **DenseNet-121** (224 px). All use ImageNet weights, focal loss ($\alpha=0.25$, $\gamma=2.0$) [5], `WeightedRandomSampler`, and Adam with cosine decay. Training: 20 epochs with early stopping on validation AUC.

4.3 BreastPairNet (Stage 2)

Shared backbone produces per-view global-average-pooled vectors $\mathbf{f}_{CC}, \mathbf{f}_{MLO} \in \mathbb{R}^d$. Two fusion heads: **concatenation** (2d-dim MLP, dropout 0.5) and **attention** (learned per-view scalar weights via softmax). The **+reg** variant adds stronger elastic deformation (16 px), mixup ($\alpha=0.4$), and label smoothing (0.05). Training: backbone frozen 3 epochs, then discriminative LR ($1e-5$ backbone, $1e-4$ head). Figure 2 illustrates the

architecture.

4.4 CrossView Transformer (Stage 3)

Replaces global-pooled fusion with spatial cross-attention. Both views pass through the shared CNN backbone to the last spatial feature map; spatial tensors are flattened to tokens and entered into a two-layer Transformer encoder where queries from each view attend to keys/values of the other. Cross-attended outputs are global-averaged and concatenated for the final classifier. Figure 3 contrasts this with BreastPairNet.

4.5 Bounding-box annotation (training only)

Ground-truth boxes (VinDr-Mammo) were drawn as white outlines (40-px line width) on training images only. For RSNA 2022, a YOLOv8s detector pre-trained on VinDr-Mammo was used (0.8% sensitivity — noisy hints only). Test sets always used original unannotated PNGs to prevent label leakage.

4.6 MC Dropout (Stage 4)

Dropout enabled at inference; BatchNorm in eval mode. $N = 20$ stochastic passes per test pair; mean probability $\bar{p}$ and binary entropy $H(\bar{p}) = -\bar{p} \log \bar{p} - (1 - \bar{p}) \log(1 - \bar{p})$ computed as uncertainty estimate. Threshold optimised via Youden’s J.

4.7 Hybrid triage (Verboom 2025 framework)

Cases sorted by ascending entropy; the K most-confident are routed to AI and the remainder to a radiologist (modelled as oracle or 85% sensitivity). Overall sensitivity = $(\text{TP}_{\text{AI}} + \text{cancers referred}) / \text{total cancers}$.

4.8 Training and evaluation

All models trained on NVIDIA RTX 5070 (12 GB VRAM) with `bfloat16` and AdamW (weight decay $1e-4$). Primary metric: AUROC. Also reported: F1 at Youden threshold, sensitivity, specificity, and **S@90%** (sensitivity at 90% specificity). 95% bootstrap CIs (BCa, $n=2000$ resamples) on the 2,000-pair test split (99 cancer-positive). Typical AUC CI width: ± 0.04 – 0.05 .

5 Experiments and Results

5.1 Stage 1 — Single-view CNN baselines

DenseNet-121 achieves the best AUC (0.7564) but only 24.8% sensitivity at the default threshold. All

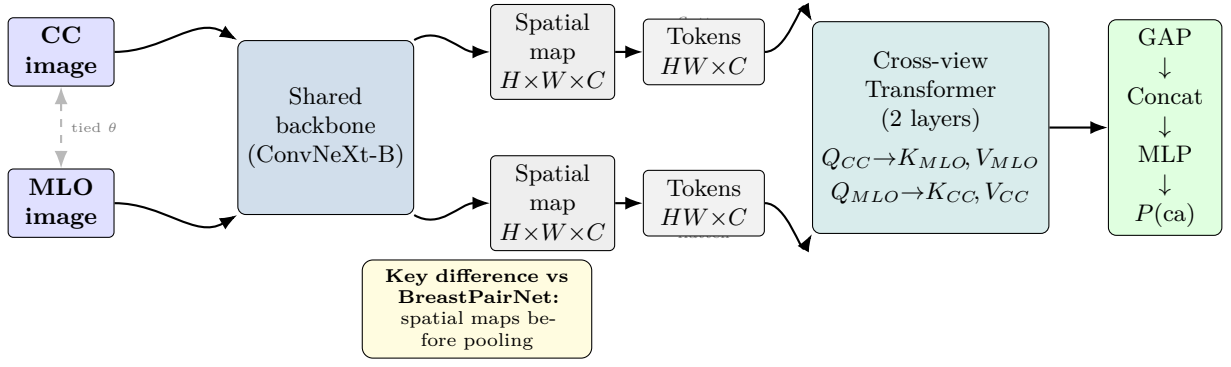


Figure 3: CrossView Transformer architecture (Stage 3). Unlike BreastPairNet, the backbone output is *not* global-averaged before fusion. Spatial feature maps are flattened to token sequences; a two-layer Transformer encoder applies bidirectional cross-attention (Q_{CC} attends to K, V from MLO and vice versa), enabling the model to localise spatially consistent findings across views before pooling. Best: ConvNeXt-Base, AUC=0.8317.

Table 3: Single-view CNN baselines on VinDr-Mammo (n=16,000 train, n=4,000 test images).

Architecture	Image size	Sensitivity	Specificity	F1	AUC	S@90%
VGG19	224 px	0.2626	0.9835	0.3323	0.7066	0.3990
ResNet-50	224 px	0.2677	0.9847	0.3430	0.7286	0.4495
EfficientNet-B3	224 px	0.2980	0.9687	0.3138	0.7330	0.3990
ResNet-50 (384 px)	384 px	0.2879	0.9821	0.3529	0.7489	0.4495
DenseNet-121	224 px	0.2475	0.9882	0.3356	0.7564	0.4444

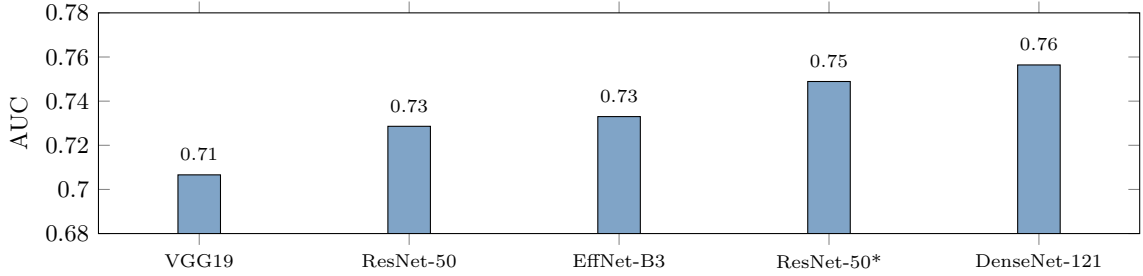


Figure 4: Stage 1: AUC by single-view architecture. ResNet-50* uses 384 px; all others 224 px.

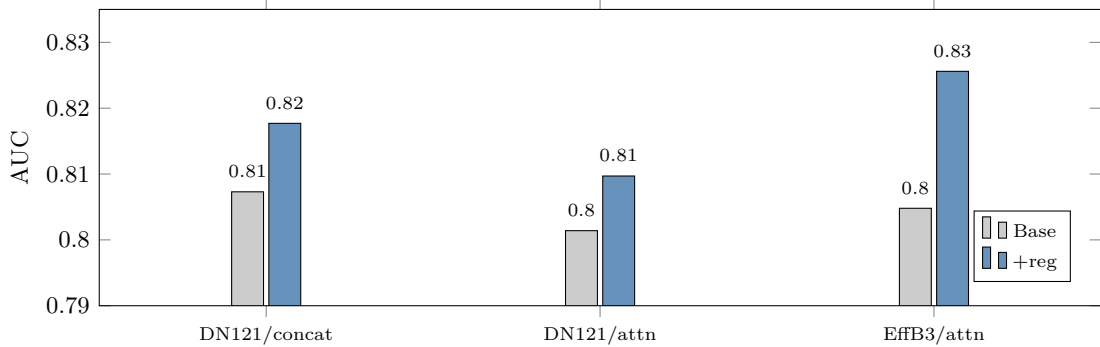


Figure 5: Stage 2: Effect of +reg on AUC for three backbone/fusion pairs.

five models overfit (train F1 $\approx$ 0.98) and reach only 40–45% sensitivity at the 90%-specificity operating point — well below the $\sim$ 85% radiologist sensitivity in screening literature [17].

5.2 Stage 2 — Multi-view BreastPairNet

The best single model is **EfficientNet-B3 + attention + +reg** at AUC=0.8256 (95% CI 0.7755–0.8707),

a +9.1% relative gain over DenseNet-121 single-view.

All point estimates in Table 4 and Table 6 are obtained by re-evaluating each best-checkpoint model on the held-out test split (n=2,000 pairs, 99 positives) at the Youden-optimal threshold, with 95% BCa bootstrap confidence intervals (n=2,000 resamples).

Three observations stand out. (i) **+reg consis-**

Table 4: BreastPairNet ablation across backbone, fusion, and resolution.

Architecture	Fusion	Image	Best AUC [95% CI]	F1	Sens	S@90% [95% CI]
ResNet-50	concat	384 px	0.7467 [0.6921–0.7998]	0.2350	0.5354	0.4646 [0.3736–0.5673]
ResNet-50	attention	224 px	0.7847 [0.7324–0.8338]	0.2522	0.5859	0.5051 [0.4023–0.6019]
ResNet-50	attention	512 px	0.7922 [0.7339–0.8429]	0.2093	0.7475	0.5152 [0.4130–0.6214]
ResNet-50	concat	224 px	0.7962 [0.7428–0.8445]	0.2363	0.6970	0.4949 [0.3958–0.5977]
DenseNet-121	attention	224 px	0.8014 [0.7487–0.8515]	0.2800	0.6364	0.5758 [0.4660–0.6667]
EfficientNet-B3	attention	224 px	0.8048 [0.7530–0.8556]	0.2138	0.7677	0.5455 [0.4476–0.6449]
DenseNet-121	concat	224 px	0.8073 [0.7577–0.8504]	0.2101	0.7374	0.5152 [0.4141–0.6100]
DenseNet-121 +reg	attention	224 px	0.8097 [0.7594–0.8591]	0.2033	0.7980	0.5556 [0.4301–0.6458]
DenseNet-121	attention	512 px	0.8150 [0.7675–0.8617]	0.1928	0.8081	0.5455 [0.4476–0.6408]
DenseNet-121 +reg	concat	224 px	0.8177 [0.7659–0.8649]	0.2357	0.7677	0.6061 [0.5047–0.7000]
EfficientNet-B3	attention	512 px	0.8175 [0.7698–0.8661]	0.2807	0.7273	0.5556 [0.4495–0.6505]
EfficientNet-B3 +reg	attention	224 px	0.8256 [0.7755–0.8707]	0.3701	0.6263	0.6263 [0.5229–0.7143]

Table 5: Isolated effect of the +reg training recipe.

Backbone	Fusion	Base AUC	+reg AUC	Gain
DenseNet-121	concat	0.8073	0.8177	+0.010
DenseNet-121	attention	0.8014	0.8097	+0.008
EfficientNet-B3	attention	0.8048	0.8256	+0.021

tently lifts AUC by +0.008–0.021 across all backbone/fusion pairs. (ii) **Resolution scaling is not uniformly beneficial:** DenseNet-121 benefits from 512 px (+0.014), but EfficientNet-B3 +reg at 224 px (0.8256) beats 512 px without +reg (0.8175). (iii) **ResNet-50 degrades at 384 px** (0.7467 vs 0.7962 at 224 px) due to batch-size reduction under VRAM pressure.

5.3 Stage 3 — CrossView Transformer

5.3.1 Performance without annotation

The best is **ConvNeXt-Base at AUC=0.8317** (+10.0% over single-view). DenseNet-121 with VinDr-Mammo+RSNA 2022 joint training reaches 0.8028, confirming incremental cross-dataset benefit. Medical pretraining (torchxrayvision) underperforms ImageNet (0.7648 vs 0.7874), suggesting domain mismatch.

5.3.2 Bounding-box annotation ablation

All four pairs show annotation hurts AUC by 0.122–0.201. At epoch 4 the backbone unfreezes and fits the white box outlines as a shortcut texture; training F1 reaches ~0.97 while test AUC collapses

because boxes are absent at test time. Remedies: (a) box-dropout regularisation or (b) test-time box injection via a lesion detector.

5.4 Stage 4 — MC Dropout uncertainty quantification

MC Dropout slightly improves AUC (+0.003) and produces a meaningful sensitivity shift (+3.0 points). More importantly, the mean entropy is **0.5670** on cancer-positive pairs versus **0.5112** on negatives — a 0.056-nat gap that enables the Stage 6 triage policy.

5.5 Stage 5 — Model ensemble

5.5.1 Original ensemble (16 BreastPairNet models)

Mean-probability ensemble of 16 BreastPairNet checkpoints combined with three breast-asymmetry models and TTA×5. **Best result in the study:** AUC=0.8458, sensitivity=0.7374, specificity=0.8296, F1=0.2944, S@90%=0.6566, S@95%=0.5657 at threshold 0.3414 (TP=73, TN=1577, FP=324, FN=26).

This ensemble does **not** include ConvNeXt-Base or

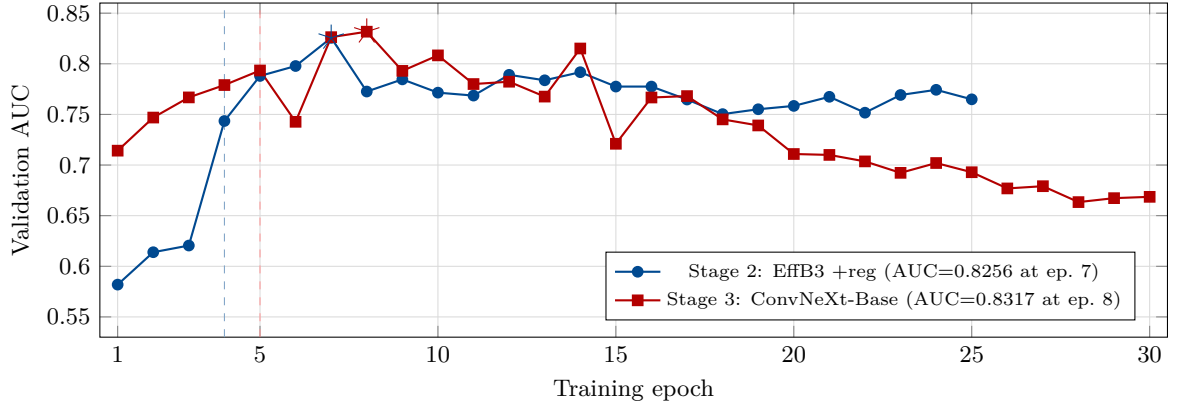


Figure 6: Validation AUC vs. training epoch for the best Stage 2 (EffB3 +reg) and Stage 3 (ConvNeXt-Base) models. Vertical dashed lines mark the backbone unfreeze point (epoch 4 for EffB3, epoch 5 for ConvNeXt). Stars (★) indicate the best-checkpoint epoch used for test-set evaluation. Both models overfit after their best epoch as training loss continues to decrease while validation AUC plateaus.

Table 6: CrossView Transformer results without bounding-box annotation.

Architecture	Training data	Best AUC [95% CI]	F1	Sens	S@90% [95% CI]
DenseNet-121 (medical init)	VinDr	0.7648 [0.7116–0.8210]	0.2374	0.5960	0.5051 [0.4020–0.6100]
EfficientNet-B3	VinDr	0.7795 [0.7276–0.8300]	0.2022	0.7273	0.4545 [0.3474–0.5500]
DenseNet-121	VinDr	0.7874 [0.7299–0.8401]	0.2678	0.6465	0.5556 [0.4474–0.6482]
DenseNet-121	VinDr+RSNA	0.8028 [0.7484–0.8536]	0.2640	0.6667	0.5354 [0.4433–0.6404]
ConvNeXt-Base	VinDr	0.8317 [0.7832–0.8753]	0.2587	0.7475	0.5758 [0.4737–0.6768]

Table 7: Bounding-box annotation ablation. Test set always uses original unannotated PNGs.

Architecture	No annotation AUC [95% CI]	With annotation AUC [95% CI]	Δ AUC
DenseNet-121	0.7874 [0.7299–0.8401]	0.5862 [0.5274–0.6433]	-0.201
DenseNet-121 +RSNA	0.8028 [0.7484–0.8536]	0.6675	-0.135
DenseNet-121 (medical)	0.7648 [0.7116–0.8210]	0.6433 [0.5886–0.6966]	-0.122
EfficientNet-B3	0.7795 [0.7276–0.8300]	0.6020 [0.5460–0.6617]	-0.178

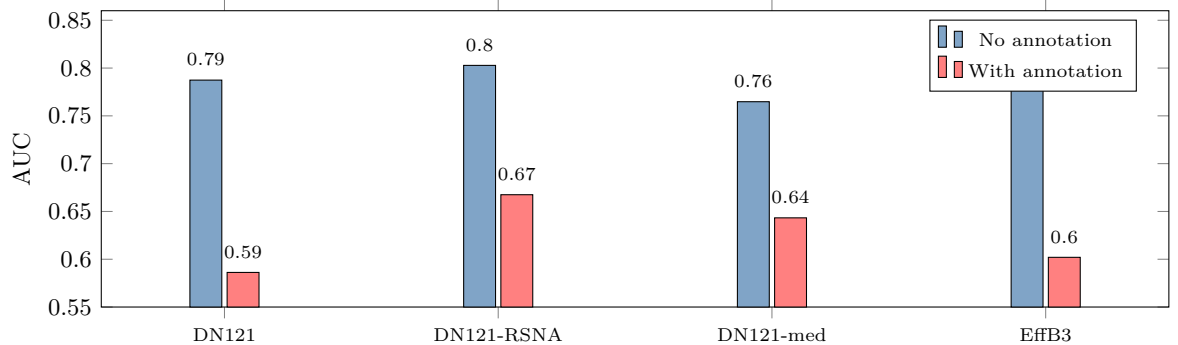


Figure 7: Stage 3b: Annotation consistently degrades AUC due to shortcut learning and test-time distribution shift.

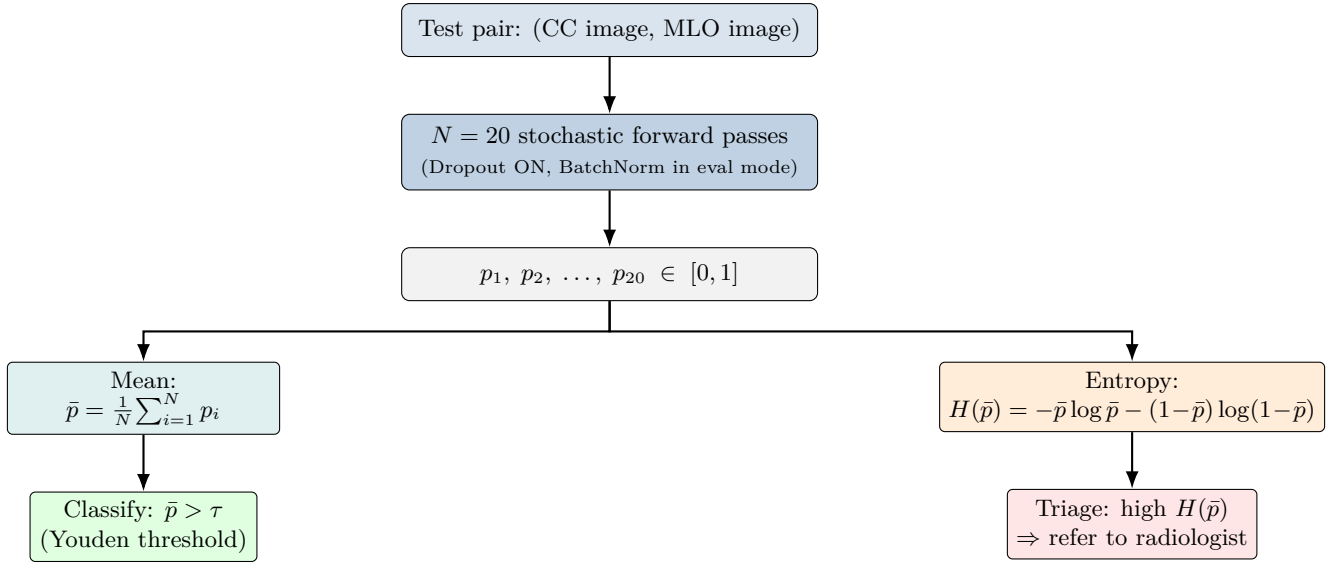


Figure 8: MC Dropout inference (Stage 4). Dropout layers remain active at test time while BatchNorm uses running statistics. Twenty stochastic passes produce a probability distribution $\{p_i\}_{i=1}^{20}$. The mean $\bar{p}$ is used for classification at the Youden-optimal threshold τ ; the binary entropy $H(\bar{p})$ ranks cases by epistemic uncertainty and routes high-uncertainty cases to a radiologist in the Stage 6 triage protocol.

Table 8: MC Dropout vs deterministic inference (DenseNet-121/concat).

Mode	AUC	Sens	Spec	S@90%	Thr.
Deterministic	0.8073	0.7374	0.7249	0.5152	0.3304
MC Drop.	0.8099	0.7677	0.7007	0.5253	0.3138

Table 9: Predictive entropy by class ($N = 20$ MC passes, 2,000 test pairs).

Entropy	Negatives ($n = 1,901$)	Positives ($n = 99$)
Mean	0.5112	0.5670
Median	0.5310	0.6221
Std dev	0.1436	0.1370
Gap: +0.056 nats. Higher entropy $\Rightarrow$ refer high-uncertainty cases.		

the same $K=50\%$ yields 77.3% — below the clinical floor. Achieving $\geq 80\%$ overall sensitivity under realistic conditions requires $K \leq 40\%$.

5.7 Comparison to literature

‡MamT4 excludes BI-RADS 3; our pipeline retains it as negative. McKinney et al. [11] (AUC=0.889) and Yoon et al. [4] meta-analysis (pooled AUC=0.87) use different datasets and are not directly comparable.

5.8 External validation on RSNA 2022

Three factors drive the gap: (1) Vietnamese vs US breast density distribution and lesion prevalence; (2) different FFDM equipment and post-processing; (3) the S@90%spec drop (-75.1%) is particularly stark at the clinical operating point, indicating the conservative side of the decision surface is most affected by domain shift. Table 17 summarises the best-performing model family at each stage.

EfficientNet-B3 +reg models.

5.5.2 Updated ensemble (25 checkpoints)

Adding all CrossView checkpoints (including lower-epoch snapshots) gives AUC=0.8441, S@90%=0.6566 (unchanged), sensitivity=0.7677. The marginal AUC drop confirms that weaker models drag the mean under mean combination; S@90% is robust.

5.5.3 Selective ensemble (top-3 +reg + top-5 CrossView + 3 asymmetry)

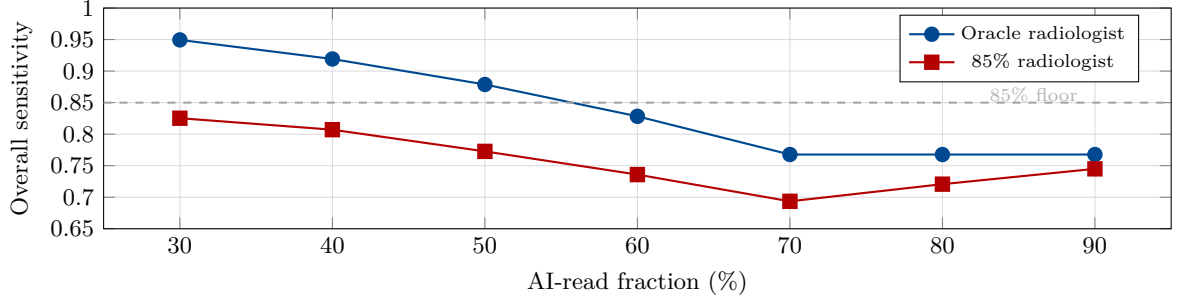
11 models, TTA $\times 5$: AUC=0.8446, S@90%=0.6667 (+0.010 over baseline). Marginally better at the clinical operating point; difference is within sampling noise.

5.6 Stage 6 — Hybrid triage simulation

Under oracle assumptions, **K=50% delivers 87.9% overall sensitivity with 50% workload reduction**. Under realistic 85% radiologist sensitivity [17],

Table 10: Best ensemble result (mean probability, TTA $\times$ 5).

Method	AUC	Sens	Spec	F1	S@90%	S@95%	Threshold
Ensemble (N=16, TTA=5)	0.8458	0.7374	0.8296	0.2944	0.6566	0.5657	0.3414

**Figure 9:** Stage 6: Overall sensitivity vs. AI-read fraction. Dashed line = 85% clinical sensitivity floor.**Table 11:** Confusion matrix at Youden threshold ($t = 0.3414$).

	Pred. negative	Pred. positive
Actual negative	TN = 1,577	FP = 324
Actual positive	FN = 26	TP = 73

Table 12: Hybrid triage sweep (oracle vs. 85% radiologist sensitivity).

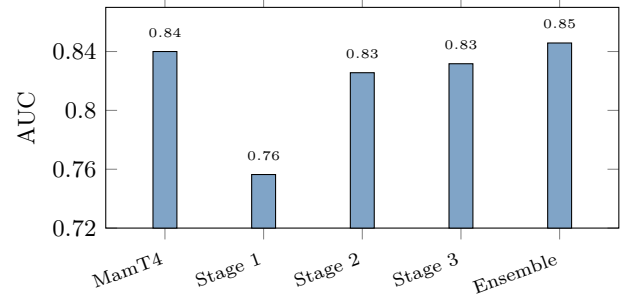
Frac	Wkld Δ	Sen (or.)	Sen (85%r)	AI Sen	Recall
30%	-30%	0.9495	0.8253	0.7059	0.0017
40%	-40%	0.9192	0.8071	0.6800	0.0013
50%	-50%	0.8788	0.7727	0.5862	0.0031
60%	-60%	0.8283	0.7359	0.5526	0.0052
70%	-70%	0.7677	0.6935	0.5400	0.0133
80%	-80%	0.7677	0.7207	0.6618	0.1305
90%	-90%	0.7677	0.7450	0.7262	0.2238

Table 13: Max AI-read fraction by sensitivity floor.

Floor	Oracle	85% rad	Workload Δ
$\geq 90\%$	40%	n/a	-40%
$\geq 85\%$	50–60%	n/a	-50% to -60%
$\geq 80\%$	60%	40%	-40% to -60%
$\geq 77\%$	70%	50%	-50% to -70%

Table 14: AUC progression on VinDr-Mammo binary cancer detection.

Method	AUC
MamT4 [18]‡	0.840
Ours Stage 1 (DenseNet-121)	0.7564
Ours Stage 2 (BreastPairNet)	0.8256
Ours Stage 3 (CrossView)	0.8317
Ours Stage 5 (Ensemble)	0.8458

**Figure 10:** AUC progression across pipeline stages vs. MamT4 prior work.**Table 15:** External validation: VinDr-Mammo (2,000 pairs, 99 cancers) vs. RSNA 2022 (4,764 pairs, 110 cancers).

Dataset	AUC [95% CI]	S@90%	Sens	Spec
VinDr-Mammo	0.8458	0.6566	0.7374	0.8296
RSNA 2022	0.6008 [0.5577–0.6585]	0.1636	0.3909	0.7598

Table 16: Performance drop: VinDr-Mammo $\rightarrow$ RSNA 2022.

Metric	VinDr-Mammo	RSNA 2022	Drop	Rel.
AUC	0.8458	0.6008	-0.245	-29%
S@90%spec	0.6566	0.1636	-0.493	-75%
Sensitivity	0.7374	0.3909	-0.347	-47%
Specificity	0.8296	0.7598	-0.070	-8%

Table 17: Summary of the best-performing model family in each stage.

Family	Best AUC	95% CI	S@90%	Key advantage
Single-view CNN	0.7564	—	0.4444	Simple baseline
BreastPairNet (concat)	0.8177	0.7659–0.8649	0.6061	Robust across backbones
BreastPairNet (attention)	0.8256	0.7755–0.8707	0.6263	Best regularised
CrossView Transformer	0.8317	0.7832–0.8753	0.5758	Spatial alignment
Ensemble (16+3, TTA×5)	0.8458	—	0.6566	Best overall ★

6 Discussion

Why multi-view helps. The +9.1% gain from BreastPairNet is consistent with the radiological reality that lesions visible in CC may be ambiguous in MLO and vice versa. Joint conditioning removes view-specific false positives.

Why +reg helps. With only ~395 cancer-positive training pairs, stronger augmentation (elastic deformation, mixup, label smoothing) is a powerful implicit regulariser (+0.008–0.021 AUC across all tested variants).

Why CrossView edges out BreastPairNet. ConvNeXt-Base (0.8317) vs EfficientNet-B3 +reg (0.8256): $\Delta=0.006$, within the ± 0.04 CI. Exchanging spatial feature maps before global pooling preserves localisation information that BreastPairNet’s independent pooling discards.

Why medical pretraining hurt. Chest-X-ray features (lung parenchyma, mediastinum, rib edges) are domain-mismatched to mammographic texture. ImageNet’s broader texture vocabulary provides a better starting point (0.7874 vs 0.7648).

Why annotation always hurt. Painted boxes provide a strong shortcut perfectly correlated with the cancer label in training, but absent at test time. Remedies: box-dropout regularisation or test-time box injection.

MC Dropout. The +0.003 AUC gain is within the 95% CI. The real value is the sensitivity boost (+3.0 points) and the 0.056-nat entropy gap between classes, which enables triage.

Triage simulation. Under realistic 85% radiologist sensitivity, $K=50\%$ yields only 77.3% overall sensitivity — 10.6 points below the oracle result. $K \leq 40\%$ is required to stay above the 80% clinical floor in non-oracle conditions.

Confidence intervals. With 99 positive test pairs, AUC CI width is ± 0.04 –0.05. Differences < 0.01 AUC are within sampling noise. BreastPairNet (0.8256) and CrossView (0.8317) are not significantly separated.

Limitations. (i) Small positive test count (99 pairs) limits fine-grained comparison; (ii) large domain gap

to US FFDM (AUC=0.6008); (iii) MamT4 comparison uses a slightly different BI-RADS exclusion; (iv) 7,054 non-CC/MLO RSNA images excluded from pair construction.

Future work. (1) Box-dropout or test-time box injection; (2) CrossView at 512px with gradient checkpointing; (3) Validate on CBIS-DDSM and INbreast; (4) Tighter ensemble of +reg BreastPairNet + best CrossView + asymmetry; (5) Fine-tune on RSNA training pairs; (6) Prospective triage study with real reader sensitivity data.

7 Conclusion

We have presented a six-stage pipeline for breast cancer detection on VinDr-Mammo: **Stage 1** (DenseNet-121, AUC=0.7564) → **Stage 2** (BreastPairNet +reg, AUC=0.8256, +9.1%) → **Stage 3** (CrossView Transformer, AUC=0.8317, +10.0%) → **Stage 5** (16-model ensemble, AUC=0.8458, +11.8%). Bounding-box annotations consistently degraded AUC by 0.12–0.20 due to shortcut learning — the most actionable negative finding. MC Dropout produced a 0.056-nat entropy gap enabling hybrid triage: 50% AI reads preserves 87.9% sensitivity under oracle conditions, but only 77.3% under realistic 85% radiologist sensitivity, requiring $K \leq 40\%$ to stay above the clinical floor. External validation (RSNA AUC=0.6008) reveals that domain generalisation to US FFDM remains a critical open challenge.

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